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Sl. No.	Course Code	Course	L	T	P	Evaluation Scheme		Total	Credits	Course Type
						CIA	ESE			
1	BCA30104	Data Structure and Algorithm	3	1	0	40	60	100	4	Major
2	BCA39105	Data Structure and Algorithm Lab	0	0	4	40	60	100	2	Major
3	BCA39106	Python Lab	0	0	4	40	60	100	2	Major
4	BCA30203	Artificial Intelligence	3	1	0	40	60	100	4	Minor
5	BCA30001	Vedic Mathematics	3	0	0	40	60	100	3	IC
6	AEC00006	Modern Indian Language A. Hindi B. Bengali C. Alternative English D. MOOC	2	0	0	40	60	100	2	AEC
7	BCA37002	Internet and Web Technologies	2	0	2	40	60	100	3	SEC
TOTAL			**13**	**2**	**10**	**280**	**420**	**700**	**20**	

*MOOC/Equivalent MOOC approved by BoS Chair (Evaluation scheme as per Blended Learning and MOOCs Policy)

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Course Code: BCA30104
Course Name: Data Structure and Algorithm
Weekly Contact Hours: 3L + 1T
Credit: 4
Total Allotted Hours: 45L + 15T

Course Objective: This course is designed to help the students understand data structures, algorithm design method and systematic way of solving problems.

Pre-requisite(s): Basic knowledge of programming.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define the basic concept of algorithm, complexity, Big O notations, and time-space trade off and analyze algorithms using asymptotic notations.
CO2: Apply different searching and sorting methods and compare them with respect to time and space complexity.
CO3: Define and apply data structures such as Stacks, Queues, Arrays, and Linked List and write algorithms to solve several problems.
CO4: Understand & implement the various non-liner data structure.
CO5: Apply and analyze graph algorithms to solve the various problems.

Module 1: Introduction **[6H]**

Introduction to Algorithms: definition and characteristics of an algorithm, Abstract Data Type (ADT), type of data structures, data structure operations, asymptotic notations – Big O, Big ω and Big θ. Recursion: definition, tail recursion, recursive functions. Arrays: basic concepts. Operations on array.

Module 2: Sorting and Searching **[12H]**

Introduction, bubble sort, insertion sort, selection sort, merge sort, shell sort, radix sort and their complexity analysis. Introduction to linear search, binary search, comparisons of searching techniques. Linked list: basic concepts, definition and representation of linked list, types of linked lists: singly linked list, doubly liked list, circular linked list, representation of Linked list in memory, operations on linked lists.

Module 3: Stack and Queue **[9H]**

Stacks: basic concepts, definition and operations on stacks, applications of stacks, infix, postfix and prefix notations, conversion from infix to postfix, evaluation of postfix expression. Queue: basic concepts, definition and types of queues - simple queues, circular queues, basic idea of double ended queue and priority queue, linked representation of stack and queue.

Module 4: Tree **[9H]**

Definition, basic concepts of tree terminologies, binary tree: memory representations, type of binary trees – strictly binary tree, complete binary tree, binary search tree. traversal of binary tree- preorder, inorder and postorder traversal. AVL Tree, B tree.

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Module 5: Graph **[9H]**

Definition, terminology and representations of graph, directed graphs, sequential representations of graphs, adjacency matrices. Graph traversal: Depth First Search (DFS) and Breadth First Search (BFS). Hashing: hash function, types of hash functions, collision, collision resolution techniques – open addressing, chaining.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1 ,K2, K4
2.	CO2	M1, M2	K3,K4, K5
3.	CO3	M3	K1,K3,K4
4.	CO4	M4	K2,K3
5.	CO5	M4, M5	K3, K4

Text books:

1. Data Structure through C Language, Samiran Chattopadhyay, Debabrata Ghosh Dastidar, Matangini Chattopadhyay, BPB Publications, 2012.

Reference books:

1. Data Structures, Seymour Lipschutz, Tata McGraw- Hill Publishing Company Limited, Schaum's Outlines.

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Course Code: BCA39105
Course Name: Data Structure and Algorithm Lab
Weekly Contact Hours: 4P
Credit: 2
Total Allotted Hours: 60P

Course Objective: The course is design to explain fundamentals of data structures and their applications essential for programming/problem solving. It will help the students understand and illustrate linear representation of data structures i.e. Stack, Queues, Lists, Trees and Graphs by sorting and searching algorithms.

Pre-requisite(s): Basic knowledge of programming.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Evaluate algorithms using arrays, recursive functions and solve different searching and sorting problems.
CO2: Use linear data structures like stack, queue, linked lists and write programs in efficient way.
CO3: Develop and implement algorithms using tree and graph to solve real life problems.

Module 1: Introduction **[8H]**

Implement array and recursion.

Module 2: Sorting and Searching **[16H]**

Implement linear search, binary search, bubble sort, insertion sort, selection sort, quick sort, merge sort, shell sort and radix sort algorithms.

Module 3: Stack and Queue **[20H]**

Implement stack and queue and their operations, check parenthesis and prefix, infix and postfix operations using own stack. Implement different insertion, deletion and traversal operations on singly, double and circular linked lists.

Module 4: Tree **[10H]**

Write programs to perform the following: a. Create a binary search tree. b. Traverse the above binary search tree recursively in pre-order, post-order and in-order. c. Count the number of nodes in the binary search tree.

Module 5: Graph **[6H]**

Write programs to implement the following graph traversal algorithms: a. Depth first search. b. Breadth first search.

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Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1, M2	K5, K6
2.	CO2	M3	K4, K5
3.	CO3	M4, M5	K4, K5, K6

Text books:

1. Fundamentals of Data Structures, Horowitz and Sahani, Galgotia 7th Edition.

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Course Code: BCA39106
Course Name: Python Lab
Weekly Contact Hours: 4P
Credit: 2
Total Allotted Hours: 60P

Course Objective: The course is designed to provide knowledge of python language. Students will be able to develop logics which will help them to create applications in Python. Also by learning the basic programming constructs they can easily switch over to any other language in future.

Pre-requisite(s): Basic Knowledge of programming.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Write programs using variables, identifiers, operators, type conversion, decision making and loops to solve various problems.
CO2: Compare list, tuple, dictionary, related operations and functions to develop algorithms for various problems.
CO3: Evaluate and criticize the concept of file handling, OOP and different packages with different types of programs in python.

Module 1: Basic Programming Constructs **[16H]**

Introduction to Programming, choosing Python, setting up Python environment, Python IDEs. Writing programs on input/output, comments, variables, data types, typecasting, arithmetic, assignment, comparison, logical, identity, membership, bitwise operators, if, if-else, if-elif-else, loop, for loop, while loop and their applications, break, continue, pass, range.

Module 2: List, Tuple, Dictionary **[12H]**

Understanding lists and writing program using list. Tuple, dictionary, their operations, creating, formatting, indexing, slicing, String methods.

Module 3: Function & Modules **[12H]**

Writing programs on function creating, calling, arguments, variables, recursion, lambda functions. Date, time, datetime modules.

Module 4: File Handling, OOPS & Event Driven Programming **[12H]**

Basic concept of file handling and I/O, programs on reading and writing to and from a file. Introduction to Class, Objects and Inheritance. Basic idea of event driven programming and GUI Programming.

Module 5: Packages **[8H]**

Basic idea of NumPy, Scipy, Matplotlib and their applications.

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Course Outcomes (COs) and Modules Mapping

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K5, K6
2.	CO2	M2,M3	K4, K5, K6
3.	CO3	M4,M5	K5

Text book:

1. Python Cookbook: Recipes for Mastering Python.

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Course Code: BCA30203
Course Name: Artificial Intelligence
Weekly Contact Hours: 3L + 1T
Credit: 4
Total Allotted Hours: 45L + 15T

Course Objective: The aim of this course is to apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.

Pre-requisite(s): Knowledge of Discrete Mathematics, Algorithm and Data Structures.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define Artificial Intelligence including topics, branches, and applications and describe Knowledge representations.
CO2: Explain and analyze the significance of intelligent agents in the Artificial Intelligence.
CO3: Compare different reasoning methods to evaluate Bayesian networks and other associations and rule mining and explain game planning approach.
CO4: Illustrate and explain the natural language processing in Artificial Intelligence.
CO5: Examine the principles, components, operations and justify other technological advancements in Robotics.

Module 1: Introduction to Artificial Intelligence **[8H]**

Definition of artificial intelligence, timelines of artificial intelligence, branches of artificial intelligence, applications of artificial intelligence, intelligent agents - structure, types of agents, environment, autonomous agents, problem solving production systems, state space representation.

Module 2: Knowledge Representation Scheme and Reasoning Concept **[15H]**

Knowledge representation - knowledge management, types of knowledge, knowledge representation-bases and structures - first order logic, unification algorithm, frames, conceptual dependency, scripts, semantic network. Reasoning - types of reasoning, non-monotonic reasoning, reasoning with fuzzy logic, rule-based reasoning, case-based reasoning, model-based reasoning systems. - Bayes rule, Bayesian networks, probabilistic inference, sample applications.

Module 3: Game Playing and Minima Procedure **[8H]**

Game playing - minimax procedure, alpha-beta pruning, combined approach, iterative deepening.

Module 4: Overview of NLP **[4H]**

Introduction, syntactic processing, semantic analysis, discourse and pragmatic processing.

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Module 5: Expert System and Robotics **[10H]**

Expert systems - characteristics, components, expert system development, knowledge engineering, application of expert systems. case studies in expert systems. Robotics – principles and components, operations, mathematics of robotics, line flow robot, applications of robots, case studies in robotics.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K2, K4, K5
3.	CO3	M2, M3	K4, K5
4.	CO4	M4	K2, K4, K5
5.	CO5	M5	K4, K6

Text books:

1. Artificial Intelligence, Ritch & Knight, TMH, 2nd Edition, 1991.

Reference books:

1. Pattern Recognition and Machine Learning, C. M. Bishop, Springer, 2006.

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Course Code: BCA30001
Course Name: Vedic Mathematics
Weekly Contact Hours: 3L
Credit: 3
Total Allotted Hours: 45L

Course Objective: Foster the love for mathematics by creating a positive attitude through Vedic and Ancient Indian Mathematics, help students appreciate ancient Indian Mathematics and its contribution to the world, develop conceptual knowledge of mathematical concepts.

Pre-requisite(s): Basic knowledge of Mathematics.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Develop the understanding of objectives and features of Vedic maths, understand the concept of addition and subtraction using completing the whole and from left to right.
CO2: Recognize the meaning of mathematical sutras in Sanskrit. manage to solve the multiplication using vertically and crosswise and one more than the previous one method and demonstrate multiplication by 11, 12 and 13 by using Vedic sutras of multiplication.
CO3: Distinguish between squaring numbers ending in 5 and squaring numbers near number 50 and manage to simplify algebraic squaring and evaluate LCF and HCF.
CO4: Apply reverse squaring to find square root of number ending in 5 and manage to solve the square root of perfect square.
CO5: Identify cube and cube roots, understand and apply division by 9 and summarize the concept of division by using straight division.

Module 1: Introduction to Vedic Mathematics **[6H]**

Contribution of Indian Mathematicians in light of arithmetic addition, history of Vedic maths, why Vedic maths, salient features of Vedic maths, Vedic maths formulas, 16 sutras, 13 sub sutras, terms and operations. High-speed addition by using the concept of computing the whole and from left to right, super-fast subtraction by Nikhilam Sutram from basis 100,1000,10,000.

Module 2: Multiplication **[14H]**

Ekadhikena Purvena method (multiplication of two numbers of two digits), Ekanyunena Purvena method (multiplication of two numbers of three digits), Urdhva Tiryagbyham method - multiplication of two numbers of three digits, Nikhilam Navtashcaramam Dashatah (multiplication of two numbers of three digits.

Module 3: Division and Divisibility **[10H]**

Decimals and fractions - Nikhilam Navtashcaramam Dashatah by two digits' divisor, Paraavartya Yojayet method by three digits' divisor, Ekadhikena Purvena method by two digits' divisor, Ekanyunena Purvena method by two digits' divisor.

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Module 4: Power and Root **[10H]**

Power: square (two-digit numbers), cube (two-digit numbers).
Root: square root (four-digit numbers), cube root (six-digit numbers).

Module 5: LCM and HCF **[5H]**

Properties of L.C.M, different methods for finding L.C.M-by listing multiples, by prime factorization, by division method, trick to find L.C.M., properties of H.C.F., method for finding H.C.F for two and three numbers-by prime factorization, by division method, trick to find H.C.F.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K3
2.	CO2	M2	K1, K3, K6
3.	CO3	M4, M5	K2, K4, K5
4.	CO4	M4	K3, K6
5.	CO5	M3, M4	K1, K2, K3, K5

Text Books:

1. Vedic Mathematics, Sri Bharatikrishna Tirthaji, Motilal Banarsidass, 1965.ISBN 81-208-0163-6.
2. Vedic Ganita, Vihangama Drishti-1, Siksha Sanskriti Uthana Nyasa, New Delhi.
3. Vedic Ganita Praneta, Siksha Sanskriti Uthana Nyasa, New Delhi.
4. Vedic Mathematics, Past, Present and Future, Siksha Sanskriti Uthana Nyasa, New Delhi.

Reference Books:

1. Discover Vedic Mathematics, Williams K.R. Vedic Mathematics Research Group, 1984. ISBN 1- 869932-01-3.
2. The Cosmic Calculator, Wiliams K.R. and M.Gaskell, Motilal Banarsidass ,2002. ISBN 81-208- 1871 -7.
3. Vertically and Crosswise, Williams,J. Pickles, Inspiration books, 1984. ISBN 1-902517-03-2.

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Course Code: AEC00006A
Course Name: हिंदी (Hindi)
Weekly Contact Hours: 2L
Credit: 2
Total Allotted Hours: 30L

पाठ्यक्रम का उद्देश्य: इस पाठ्यक्रम का उद्देश्य छात्रों में हिंदी के सामाजिक और सांस्कृतिक परिप्रेक्ष्य को समझते हुए अपने विचार को हिंदी में प्रभावी तरीके से लिखना और व्यक्त करना है। इसके अतिरिक्त यह पाठ्यक्रम छात्रों में सामाजिक विविधता को समझने तथा विभिन्न क्षेत्रों में रोजगार पाने की क्षमता विकसित करने में भी मदद करेगा।

Course Objective: The objective of this course is to help the students write and express their thoughts effectively in Hindi while understanding the social and cultural perspectives of Hindi. Apart from this, this course also helps in understanding social diversity and enables them to be employed in various sectors.

पूर्व-अपेक्षा (Pre-requisite(s)): कुछ नहीं (Nil)

Course Outcomes (COs): इस पाठ्यक्रम के अंत में छात्र (After completion of the course students will be able to:)

CO1: हिंदी भाषा और हिंदी साहित्य की उत्पत्ति और विकास को समझकर उसके परिप्रेक्ष्य को जान सकेंगे ।
(would be able to understand the origin and development of the Hindi language and Hindi literature to know its perspectives.)

CO2:राष्ट्रीयता और बहुसांस्कृतिक भारतीय समाज के मूल्य के लिए हिंदी भाषा और हिंदी साहित्य के महत्व को समझने और लागू करने में सक्षम होंगे।
(would be able to understand and implement the importance of the Hindi language and Hindi literature for nationalism and the value of multicultural Indian society.)

CO3: वाक्य निर्माण के लिए हिंदी शब्दावली और बुनियादी व्याकरण को समझने और उपयोग करने में सक्षम होंगे ।
(would be able to understand and use the Hindi vocabulary and basic grammar for sentence construction.)

CO4: हिंदी में प्रशासनिक शब्दों को समझने और आधिकारिक संचार के लिए उनका उपयोग करने में सक्षम होंगे ।
(would be able to understand the administrative words in Hindi to use them for official communications.)

CO5: प्रभावी संचार के लिए हिंदी में संचार और रचनात्मक लेखन के प्रमुख पहलुओं को समझने और उपयोग करने में सक्षम होंगे
(would be able to understand and use the key aspects of communication and creative writing in Hindi for effective communication.)

इकाई 1: हिन्दी भाषा एवं साहित्य की पृष्ठभूमि और महत्व

(**Module 1: Background of Hindi Language and Literature)** **[5H]**

हिंदी भाषा का उद्भव; हिंदी भाषा का स्वरुप और विकास; हिंदी साहित्य का इतिहास और महत्व; आधुनिक काल में हिंदी भाषा की उपयोगिता और चुनौतियां। संविधान में हिंदी भाषा का स्थान; राष्ट्रभाषा और जन संपर्क भाषा में हिंदी का महत्व; राष्ट्रवाद में हिंदी साहित्य का महत्त्व; हिंदी साहित्य के प्रमुख लेखक और उनका योगदान। मैथिलीशरण गुप्त और रामधारी सिंह दिनकर की प्रमुख रचनाये और उनका भावार्थ।

(Origin of Hindi language, Nature and development of Hindi language, History and importance of Hindi literature, Relevance of Hindi language and literature in the contemporary world and emerging challenges, Place of Hindi language in the Constitution, Importance of Hindi as national language and public contact language, Importance of Hindi literature in nationalism, Prominent writers in Hindi and their contribution, Important poems of Mathalisharan Gupt and Ramdhari Singh Dinkar and their meaning).

इकाई 2: हिंदी भाषा का सैद्धांतिक और व्यावहारिक व्याकरण

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(**Module 2: Theoretical and functional grammar of Hindi Language)** **[5H]**

शब्द विचार : संज्ञा , सर्वनाम , क्रिया और विशेषण; शब्द रूपांतरण : लिंग, वचन, कारक, काल, वाच्य; संधि , समास और उपसर्ग; वाक्य और उसके भेद।
(Word Ideas: Sangya, Sarvnam, Kriya and Visheshan; Word transformation: gender, vachan, cases, tense, subject; Sandhi, compound and prefix; Sentence and its types)

इकाई 3: हिंदी भाषा शब्दावली और इसकी हिंदी बोलियों के प्रकार
(Module 3: Hindi Language Vocabulary and its Types of Hindi Dialects) **[6H]**

शब्द उदगम : तत्सम , तद्भव, देशज, विदेशी; हिंदी का शब्द भंडार: अनेकार्थी, समानार्थी, विपरीतार्थी, भिन्नार्थी; विभिन्न हिंदी बोलियाँ; वाक्य और उपवाक्य बनाना; सरल वाक्य से मिश्र वाक्य परिवर्तन।
(Word origin: Tatsam, Tadbhava, Native, Foreign; Hindi vocabulary: synonyms, antonyms and variants; Various Hindi dialects; Sentence and its types; Making sentences and clauses; Change from simple sentence to complex sentence)

इकाई 4: कार्यालय तथा प्रशासनिक कामकाज में हिंदी
(**Module 4: Hindi in official and administrative working)** **[8H]**

हिंदी की प्रशासनिक शब्दावली; हिंदी लेखन में वाक्य शुद्धिकरण; कार्यालय सम्बंधित पत्र लेखन; अनुवाद - इंग्लिश से हिंदी में और बंगाली से हिंदी में।
(Hindi administrative terminology, sentence correction in Hindi writing, office related letter writing, Translation-English to Hindi and Bengali to Hindi)

इकाई 5 : हिंदी में संचार कौशल
Module 5: **Communication skills in Hindi** **[6H]**

हिंदी में प्रभावी सार्वजनिक भाषण और बहस; रचनात्मक और सृजनात्मक लेखन; हिंदी में पत्राचार, निबंध , ज्ञापन और अनुस्मारक लेखन।
(Effective public speaking and debate in Hindi; creative and creative writing; Writing correspondence, essays, memos and reminders in Hindi.)

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	KI, K2
2.	CO2	M2	K1, K2, K3
3.	CO3	M3	K1, K2, K3
4.	CO4	M4	K1, K2, K3, K4
5.	CO5	M5	K1, K2, K3, K4

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पाठ्य पुस्तक (Text Books):

1. हिंदी साहित्य का इतिहास , डाक्टर नागेंद्र, राजकमल प्रकाशन
2. अनुवाद और संचार, डॉक्टर पूरनचंद टंडन , राजपाल एंड संस , दिल्ली
3. सैद्धांतिक हिंदी व्याकरण, डॉक्टर नागरत्ना राव और वैशाली सालियान, वाणी प्रकाशन, दिल्ली
4. राजभाषा हिंदी और राजकीय पत्र व्यवहार, डॉक्टर घनश्याम अग्रवाल, जय भारती प्रकाशन, प्रयागराज

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Course Code: AEC00006B
Course Name: বাংলা (Bengali)
Weekly Contact Hours: 2L
Credits: 2
Total Allotted Hours: 30L

Prerequisite(s): Ability to read and write the Bengali Language.

কোর্সের উদ্দেশ্য: এই কোর্স শিক্ষার্থীদের সঠিকভাবে বাংলা ভাষা ব্যবহার করতে সাহায্য করবে এবং সামাজিক এবং পেশাদার পরিবেশে কথোপকথন করতে সাহায্য করবে।
Course Objective: This course will help the learners to develop sufficient abilities to use Bengali language appropriately and communicate in social and professional surroundings.
প্রাথমিক প্রয়োজনীয়তা: বাংলা ভাষা এবং এর গুরুত্ব সম্পর্কে প্রাথমিক উপলব্ধি।
Prerequisite(s): Basic understanding of Bengali and its importance.

কোর্সের ফলাফল: কোর্স শেষে, ছাত্ররা সক্ষম হবে:
কোর্সের ফলাফল ১: ভাষার দৈনন্দিন ব্যবহারে ব্যাকরণগত ভাবে সঠিক বাংলা বলতে ও লিখতে।
কোর্সের ফলাফল ২: সঠিক বাক্য গঠন, প্রশ্ন করা, বাক্যের শুদ্ধতা এবং সাধারণ ত্রুটি চিহ্নিতকরণ করতে।
কোর্সের ফলাফল ৩: মানুষের চরিত্র এবং বাস্তব জীবনের পরিস্থিতি আরও ভালভাবে বিশ্লেষণ করার জন্য গল্প এবং কবিতা পড়ার মাধ্যমে পাঠ্যগুলি বোঝার ক্ষমতা উন্নত করা।
কোর্সের ফলাফল ৪:আনুষ্ঠানিক চিঠিপত্র, সংবাদ নিবন্ধ এবং অনুচ্ছেদ লেখা, ধারণা সম্প্রসারণ, বিজ্ঞাপন লেখার মাধ্যমে প্রাসঙ্গিক অভিব্যক্তি তৈরি করা।
Course Outcomes (COs): At the end of the course, students will be able to:

CO1: Speak and write grammatically correct Bengali in the everyday use of the language.
CO2: Formulate proper sentences, frame questions, spellings and identify common errors.
CO3: Comprehend texts through reading stories and poetry to analyse human characters and real life situations better.
CO4: Produce pertinent expressions through writing of formal Letters, News Articles, Paragraphs, Expanding ideas, Advertisement writings

মডিউল ১: প্রয়োগ-ভিত্তিক ব্যাকরণ ৭ ঘণ্টা

পদ পরিচয়, সন্ধি, সমাস, কারক বিভক্তি, বাচ্য, বাক্যপরিবর্তন, প্রত্যয়, ক্রিয়ার কাল ও অব্যয়।

Module 1: Application-Based Grammar [7H]

Podh Porichoy, Sondhi, Somash, Karak Bibhokti, Bachho, Bakyoporibarton, Prottoy, Kriya r Kaal o Abboy

মডিউল ২: শব্দকোষ এবং স্ব-অভিব্যক্তি ৬ঘণ্টা

বাংলা বানান রীতি, বাংলা ভাষার শব্দ অভিধান - পরিভাষা প্রয়োগ (বিজ্ঞান, সাহিত্য, ধর্ম, সংস্কৃতি, অর্থনীতি, ব্যবসায়), বাক্য রচনা, অশুদ্ধি সনাক্ত করা, ইংরেজি থেকে বাংলা অনুবাদ।

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Module 2: Vocabulary and Self-Expression **[6H]**

Bangla Banan Riti, Bangla Bhashar Sobdobhandar – Poribhasha Proyog (Bigyan, Literature, Religion, Culture, Economics, Business), Bakko Rochona, Ashuddhi Songshodhon, Engreji theke Bangla e Anubad.

মডিউল ৩: ছোট গল্প **৬ ঘণ্টা**

১) মণিহারা - রবীন্দ্রনাথ ঠাকুর
২) হারানের নাতজামাই - মানিক বন্দ্যোপাধ্যায়

Module 3: Short Stories : **[6H]**

I) Monihara – Rabindranath Thakur

II) Haran er Natjamai – Manik Bandyopadhyay

মডিউল ৪: কবিতা **৬ ঘণ্টা**
১) বনলতা সেন - জীবনানন্দ দাশ
২) আমার কৈফিয়ত - কাজী নজরুল ইসলাম

Module 4: Poems: **[6H]**

I) Bonolota Sen – Jeebonanando Das

II) Amar Koifiyat – Kazi Nazrul Islam

মডিউল ৫: লেখা দক্ষতা **৫ ঘণ্টা**

সংবাদপত্রে প্রতিবেদন রচনা, প্রাতিষ্ঠানিক পত্রলিখন, বিজ্ঞাপন রচনা, অনুচ্ছেদ রচনা, ভাবার্থ ও ভাবসম্প্রসারণ, সারাংশ।

Module 5: Writing Skills **[5H]**

Sonbadpotre Protibedon Rochona, Protisthanik Patrolikhon, Bigyapon Rochona, Anuchhed Rochona, Bhabartho o Bhabsamprosaron, Sarangsho.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K3
2.	CO2	M2	K2, K6
3.	CO3	M3	K2, K4
4.	CO4	M4, M5	K3, K6

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পাঠ্যবইসমূহ:

১। সুকুমার সেন, ২০০০, ভাষার ইতিবৃত্ত, কলকাতা: আনন্দ।
২। রমেশ্বর শ', ১৯৮৪, সাধারণ ভাষাবিজ্ঞান ও বাংলা ভাষা, কলকাতা: পুস্তকবিপণী।
৩। আব্দুল কালাম মনজুর মোরশেদ, ১৯৯৭, আধুনিক ভাষাতত্ত্ব, কলকাতা: নয়াউদ্যোগ।
৪। সুনীতি কুমার চট্টোপাধ্যায়, ২০১১, ভাষা প্রকাশ ও বাংলা ব্যাকরণ, কলকাতা: প্রকাশ ভবন।
৫। বীরেন্দ্র দত্ত, ১৯৮৫, বাংলা ছোটগল্প ও প্রসঙ্গ, কলকাতা: রত্নাবলী।
৬। বৈদ্যনাথ বন্দ্যোপাধ্যায়, ২০০৬, সেকালএকালের সংবাদ পরিবেশনের ধারা ও বিচিত্র সংবাদ, কলকাতা: আনন্দপ্রকাশন।

Text Books:

1. Sukumar Sen, 2000, *BhasharItibritto,* Kolkata: Ananda.
2. Rameshara Shaw, 1984, *Sadharanbhashabigan O banglaBhasha,* Kolkata: PustakBipani.
3. Abdul KalamManjurMorshed, 1997, AdhunikBshashatattwa, Kolkata: NayaUdyog.
4. Suniti Kumar Chattopadhayay, 2011, Bhasha Prakash O Bangla Byakaran, Kolkata: Prakash Bhaban
5. BirendraDatta, 1985, *Bangla ChatoGalpo O Prasanga*, Kolkata: Ratnaboli.
6. Baidyanath Bandopadhyay,2006,*SekalEkaler Sangbad ParibeshanerDhara O Bichitra Sangbad,*Kolkata: AnandaPrakashan.

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Course Code: AEC00006C
Course Name: Alternative English
Weekly Contact Hours: 2L
Credit: 2
Total Allotted Hours: 30L

Course Objective: This course aims to develop a solid command on the English language, enhance vocabulary, and adopt strategies for effective reading and writing, with the overall aim to enable students to succeed in their personal and professional life.

Prerequisite(s): Basic knowledge of the English language.

Course Outcomes: At the end of the course, students will be able to:

CO1: Develop proficiency in English and express themselves in the language with greater accuracy and confidence through examining and understanding messages communicated in the English language.
CO2: Recall, understand, analyse and interpret texts in relation to themes, structure and style; expand their vocabulary and enhance their linguistic competency through literary works.
CO3: Understand the importance of effective reading, explore various aspects of language comprehension, develop coherence, and competence through reading activities.
CO4: Understand the different techniques of written communication and apply the mechanics of writing to construct effective paragraphs which can help in longer compositions, summarize and expand ideas and write letters and reports.
CO5: Understand and apply their knowledge to develop competency in English communication skills and build confidence that will be useful as a life skill to evaluate various situations irrespective of their careers or fields of specialization.

Module I: Grammar and Usage **[6H]**
Tense, Prepositions and Phrasal Verbs, Subject Verb Agreement, Verb Forms, Transformation of Sentences, Common Errors.

Module II: Selected Prose **[6H]**

The Homecoming by Rabindranath Tagore.
The Thief's Story by Ruskin Bond.

Module III: Selected Poetry **[6H]**
'Night of the Scorpion' by Nissim Ezekiel.
'The Family Home' by Kamala Das.

Module IV: Reading Skills **[6H]**
Significance of reading.
Strategies of reading.
Comprehension based on a given unseen passage.

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Module V: Writing Skills **[6H]**

Paragraph writing.
Newspaper Reports.
Formal and Informal Letters.
Summarizing and paraphrasing.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels(as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1, M4, M5	KI, K2, K3
2.	CO2	M2, M3, M4	K1, K2, K3,K4
3.	CO3	M2, M3, M4	K2, K3
4.	CO4	M1, M5	K1, K3, K6
5.	CO5	M1, M4, M5	K2, K3, K5

Text books:

1. Theory and Practice of Journalism, B.N.Ahuja, Third Edition: 1988, Surjeet Publication, New Delhi.
2. Professional Journalism, M.V.Kamath, Reprint:1991, Vikas Publishing House Pvt. Ltd, New Delhi.
3. Collected Stories, Rabindranath Tagore – Rabindranath Tagore, Projapoti, 2015.

Reference books:

1. R.K.Murthi, "Journalism (A Guide to Free-Lancing)", 1996 edition, Reliance Publishing House, New Delhi.
2. M.V.Kamath, "The Journalistic Handbook", Reprint 1990, Vikas Publishing House Pvt. Ltd. New Delhi.

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Course Code: BCA37002
Course Name: Internet and Web Technologies
Weekly Contact Hours: 2L+2P
Credit: 3
Total Allotted Hours: 30L+ 30P

Course Objective: This course aims to get hands-on experience on JavaScript and jQuery. To learn how to work with binding events to the controls in JavaScript, how to download jQuery library and refer it to the Html page, selecting the Html elements by name, attribute name, id or by content, traversing of Html elements, handling different events for different controls, how to provide effects to the elements or sections in the Html page, manipulating elements by adding CSS classes dynamically, by inserting elements.

Pre-requisite(s): Basic knowledge in programming.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define the semantic structure of HTML, CSS, HTML DOM to implement in web programming.
CO2: Apply the concepts of JavaScript, test the output through suitable program.
CO3: Use JQuery to develop event and choosing valid structure as well as content.
CO4: Compare and access the DOM structure of web document and object properties.
CO5: Design and develop dynamic web pages with usage of bootstrap and server-side script writing.

Module 1: Language Syntax, HTML Forms, HTML DOM **[3H(L)]**

Variable declaration, operators, control statements, error handling, understanding arrays, function declaration, HTML document object model, working with HTML form and its elements, other Document Object Model.

Practical: **[3H(P)]**

Implementation of variable declaration, operators, control statements, error handling, understanding arrays, function declaration, HTML document object model, working with HTML form and its elements, other Document Object Model.

Module 2: Selectors: HTML Manipulation in JavaScript **[4H(L)]**

Basic Selectors, Getter and Setter values from elements, handling attributes, inserting new elements, deleting/removing elements, CSS manipulations, dimensions, positioning.

Practical: **[4H(P)]**

Implementation of Basic Selectors, Getter and Setter values from elements, handling attributes, inserting new elements, deleting/removing elements, CSS manipulations, dimensions, positioning.

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Module 3: Effects, Events in JQuery **[6H(L)]**

Showing/hiding elements, sliding elements, fading elements, deleting animation elements, custom animation working with events.

Practical: **[6H(P)]**

Working with showing/hiding elements, sliding elements, fading elements, deleting animation elements, custom animation working with events.

Module 4: Classes, Attributes & Properties Event Listeners in Vanilla JS **[9H(L)]**

Element and class List, Element.className, Element.getAttribute (), Element.setAttribute(), Element.removeAttribute(), and Element.hasAttribute().

Practical: **[9H(P)]**

Implememntation of element and class List, Element.className, Element.getAttribute (), Element.setAttribute(), Element.removeAttribute(), and Element.hasAttribute().

Module 5: Bootstrap Grid and Components **[8H(L)]**

How to apply bootstrap grid, what is bootstrap form layout, what is bootstrap button, display images in different styles like circle shape. Bootstrap dropdown menu component, button groups and button toolbar, add forms and other controls to navbar.

Practical: **[8H(P)]**

Working with bootstrap grid, what is bootstrap form layout, what is bootstrap button, display images in different styles like circle shape. Bootstrap dropdown menu component, button groups and button toolbar, add forms and other controls to navbar.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1,M2	K1,K3
2.	CO2	M2	K3, K4, K6
3.	CO3	M3	K3,K6
4.	CO4	M3, M4	K4, K5
5.	CO5	M5	K5, K6

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Text Books:

1. The complete reference HTML & CSS, Thomas A. Powell, 5th Edition.

Reference Books:

1. jQuery pocket reference, David Flanagan, O'REILLY, Ist Edition, 2010.
2. Learning jQuery, Karl Svedberg, Jonathan Chaffer, Packt, 2013.
3. HTML, CSS & Java Script web publishing,Laura Lemay,Rafe Colburn, Jennifer Kyrnin 7th Edition.